

Welcome

we will start at 9:08 pm
today

Agenda

① Docker compose

② CA.

Dockercompose



Single configuration file that describes an entire
multicontainer app & how the containers communicate
with each other.

↑ Microservice

flipkart case study

Flipkart

Team

User team

→ login service

Product team

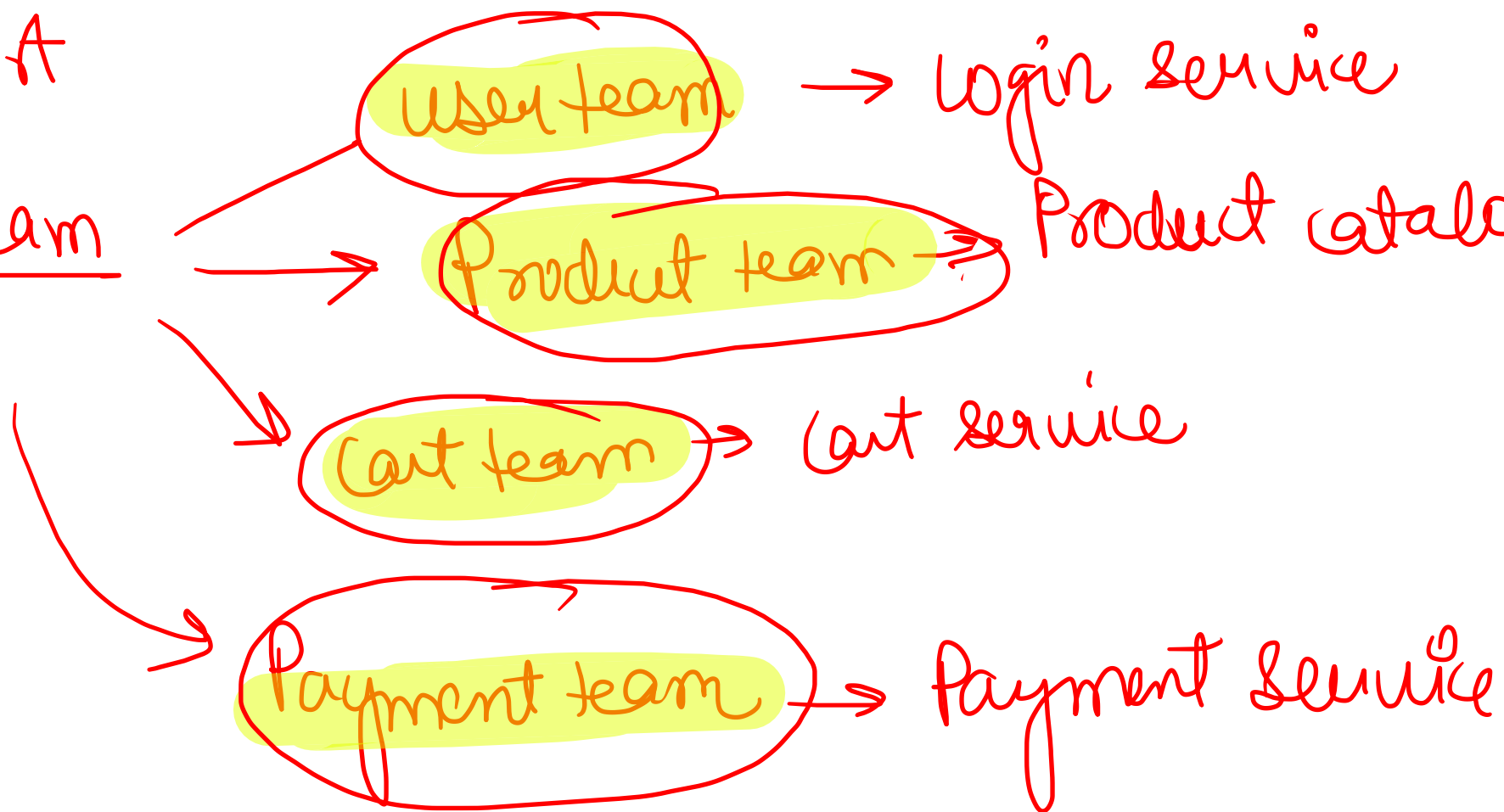
→ Product catalog service

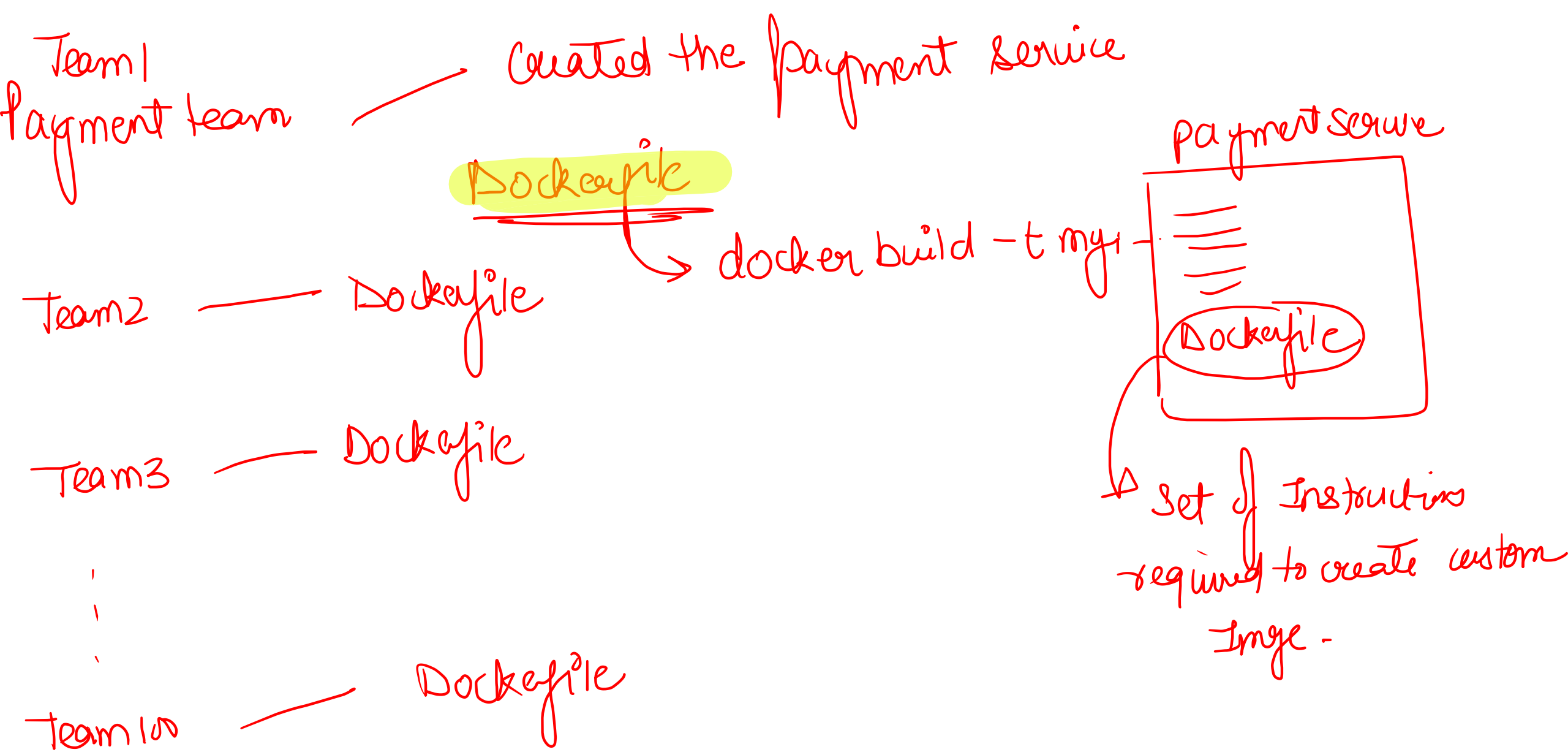
Cart team

→ Cart service

Payment team

→ Payment service





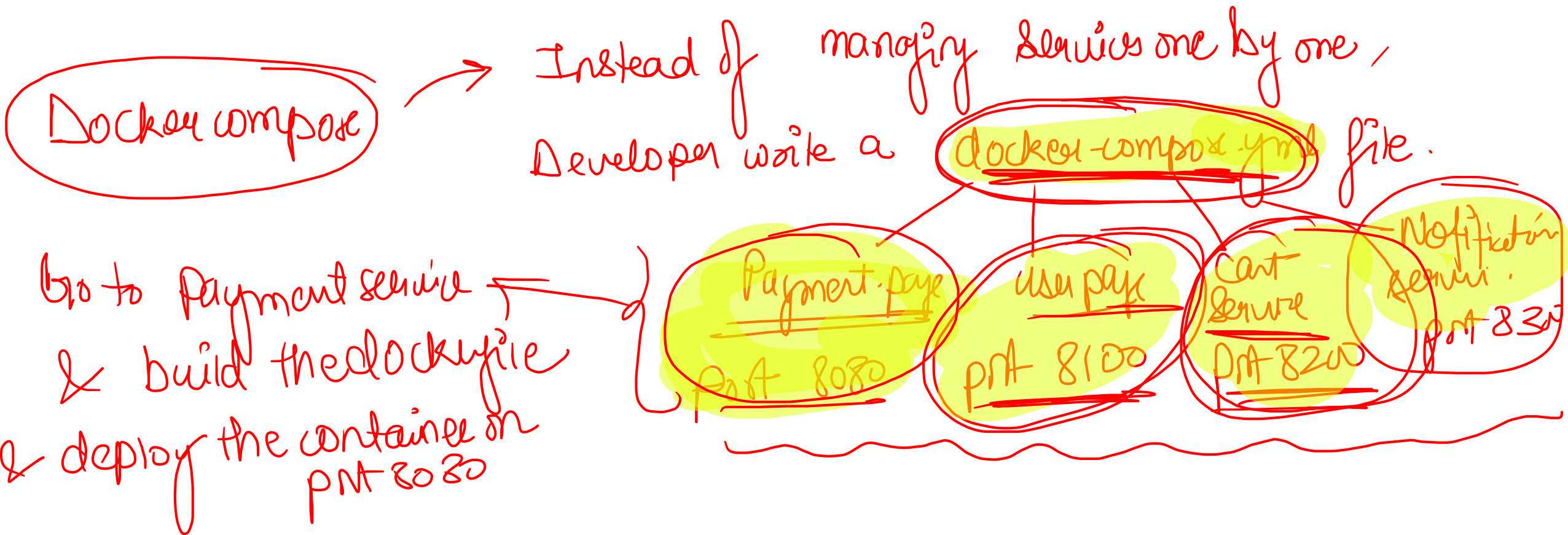
docker build -t ~~payment~~ paymentSvc:v1.

docker run -d -p 8080:8080 paymentSvc:v1

⋮
All the microservices

} DevOps

(CRAZY)



docker-compose.yml

version: '3.8'

services:

login-service

build: ./login-page

payment-service:

build: ./Payment-service

cart-service:

build: ./cart-service

docker build ./login-page.
Docker automatically looks for
login-page/Dockerfile.

Project ÷

flipkart

→ directory - Payment-service

└─ jar

└─ Dockerfile

→ directory ÷ login-page

└─ jar

└─ Dockerfile

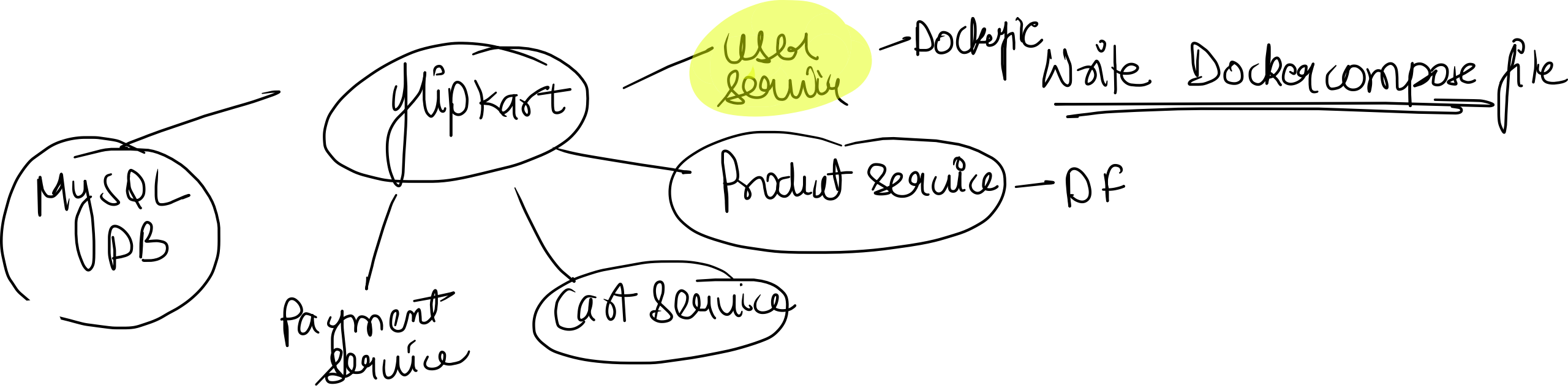
⋮

→ docker-compose.yml

→ docker compose up-d

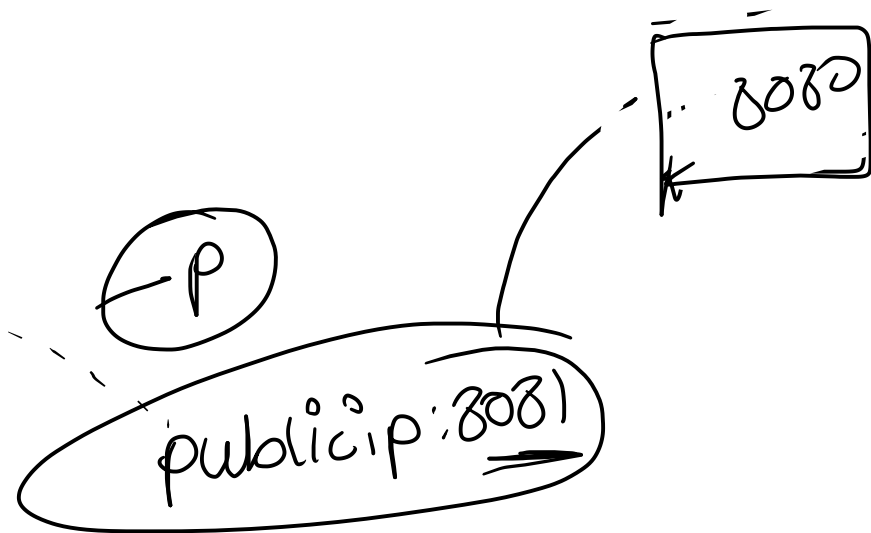
Build the dockerfiles

& deploy app in container



Services:

→ user-service:
build : . / user-service
ports :
- 8081 : 8080



→ product-service:
build : . / product-service
ports :
- 8082 : 8080

Publicip: 8082

} → Start MySQL first
Then start payment service.

→ mysql: depends-on: mysql
image: mysql:8
environment:

MYSQL-ROOT-PASSWORD: root123
MYSQL-DATABASE: flipkart

Download MySQL
Image

create database
automatically

Launch ec2 machine — ubuntu

① Sudo Su

apt update

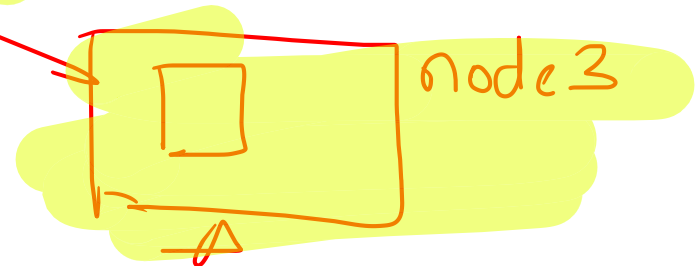
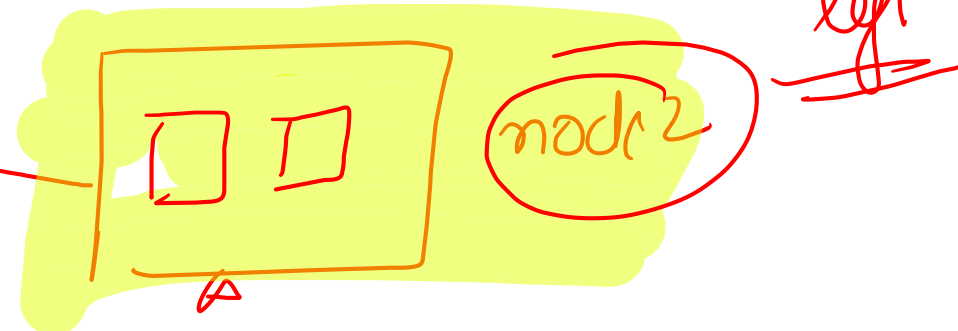
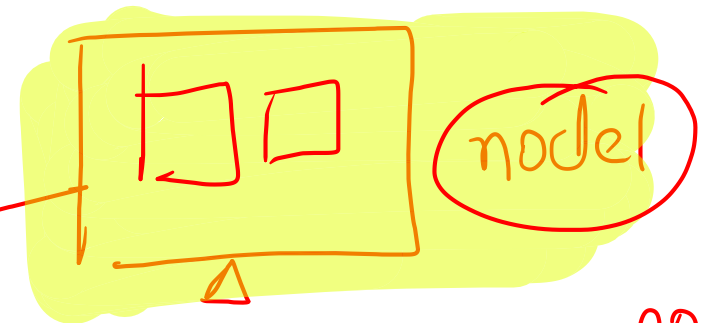
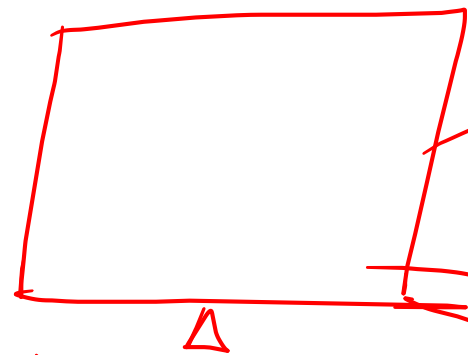
apt install docker.io -y

apt install docker-compose -y

kubernetes automatically adds or remove worker nodes based on workload demand.

when you have more traffic more workload (more nodes) master
& less traffic is less workload (less nodes)

cluster autoscaling (K8s)



CA lab

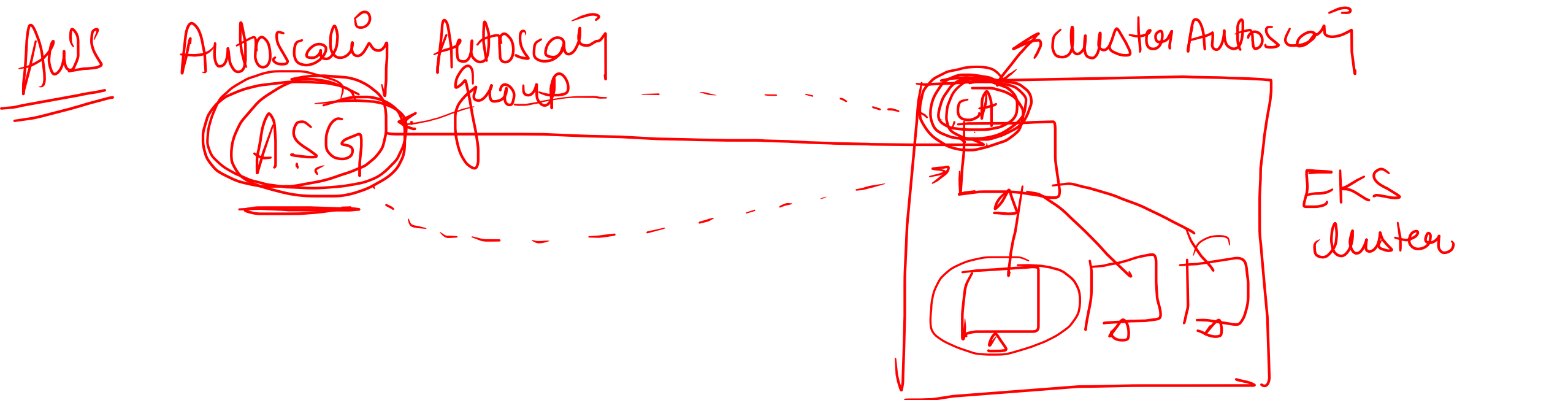
EKS cluster

Role

permission
to talk to
AWS Resource

6

↓
AWS Autoscaling
group



we will configure CA to instruct ASG to Increase or decrease the machines.

